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Studierichting: Informatica
Oefeningenreeks 2E nr. 7.2

$$\lim_{x\to 0}\left(\frac{\ln(1+x)}{5x}\right)$$
$$= \lim_{x\to 0}\left(\frac{1}{5}\cdot\frac{1}{x}\cdot\ln(1+x)\right)$$
$$= \lim_{x\to 0}\left(\frac{1}{5}\cdot\ln\left((1+x)^{\frac{1}{x}}\right)\right)$$
$$= \frac{1}{5}\cdot\ln\left(\lim_{x\to 0}(1+x)^{\frac{1}{x}}\right)$$
$$= \frac{1}{5}\cdot\ln(e)$$
$$= \frac{1}{5}\cdot 1$$
$$= \frac{1}{5}$$

References